

JOURNAL CLUB 1

Analysis of **INFLUENZA ON EXPERIMENTAL AUTOIMMUNE ENCEPHALOMYELITIS**



AGENDA

- Learn about the Differentially expressed genes & RNA seq.
- Understand findings from the study.
- Find out differentially expressed genes in the cerebellum of mice post influenza infection using R.

HYPOTHESIS

- Upper Respiratory Infections are believed to exacerbate symptoms of autoimmune diseases in the Central Nervous System.
- This was tested in the study using mice that were prone autoimmune diseases.

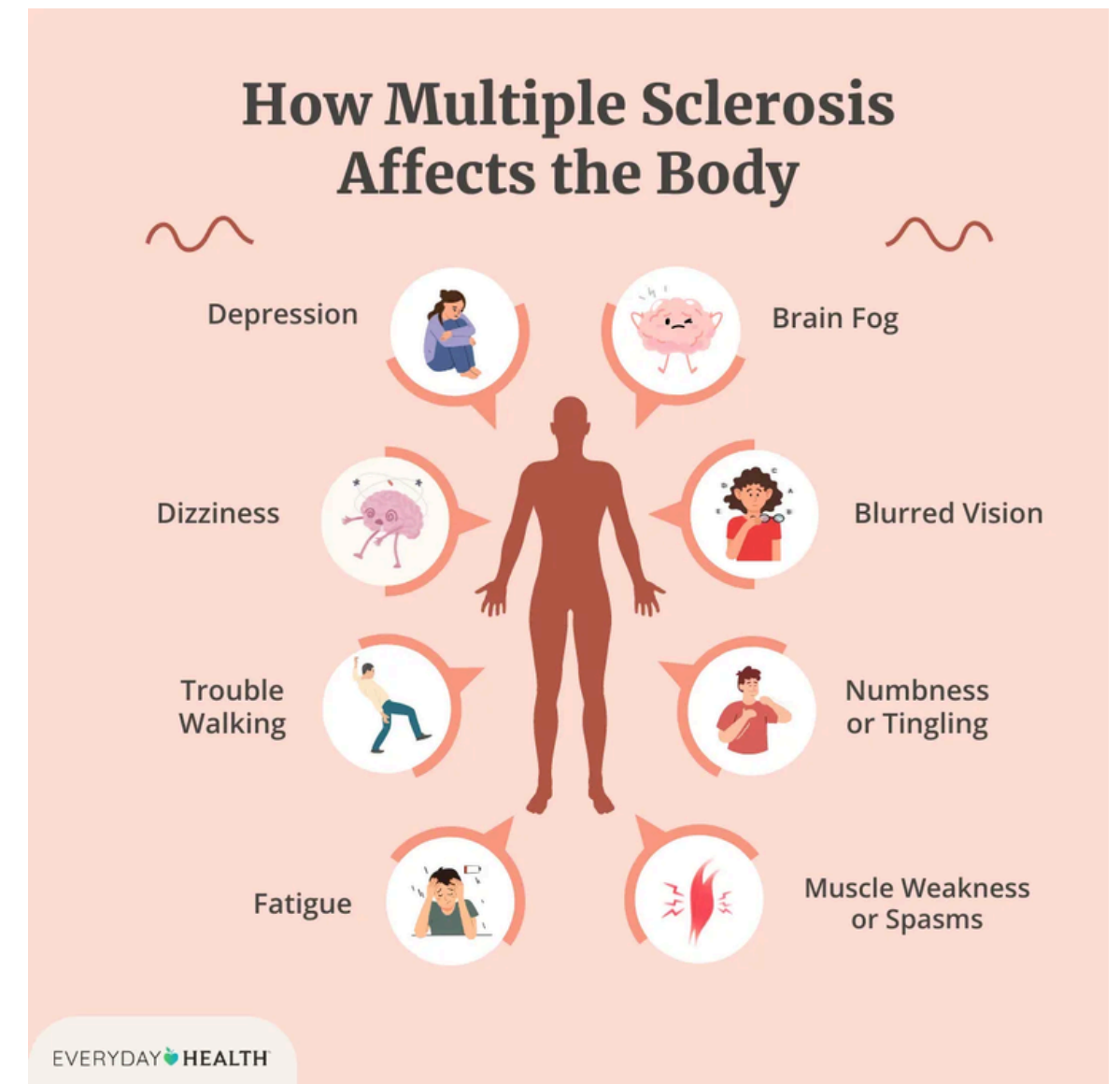
HOW DID THEY REACH THE HYPOTHESIS?

- MS patients exhibit an oscillating diseases course.
- This means periods of remission of relapse
- Surprisingly, 27-41% of MS patients will suffer exacerbated symptoms within 5 weeks post upper-respiratory infection.
- So upper-respiratory infection with viral pathogens can cause relapse, but the mechanisms remain poorly defined.

BACKGROUND

MS - Multiple Sclerosis

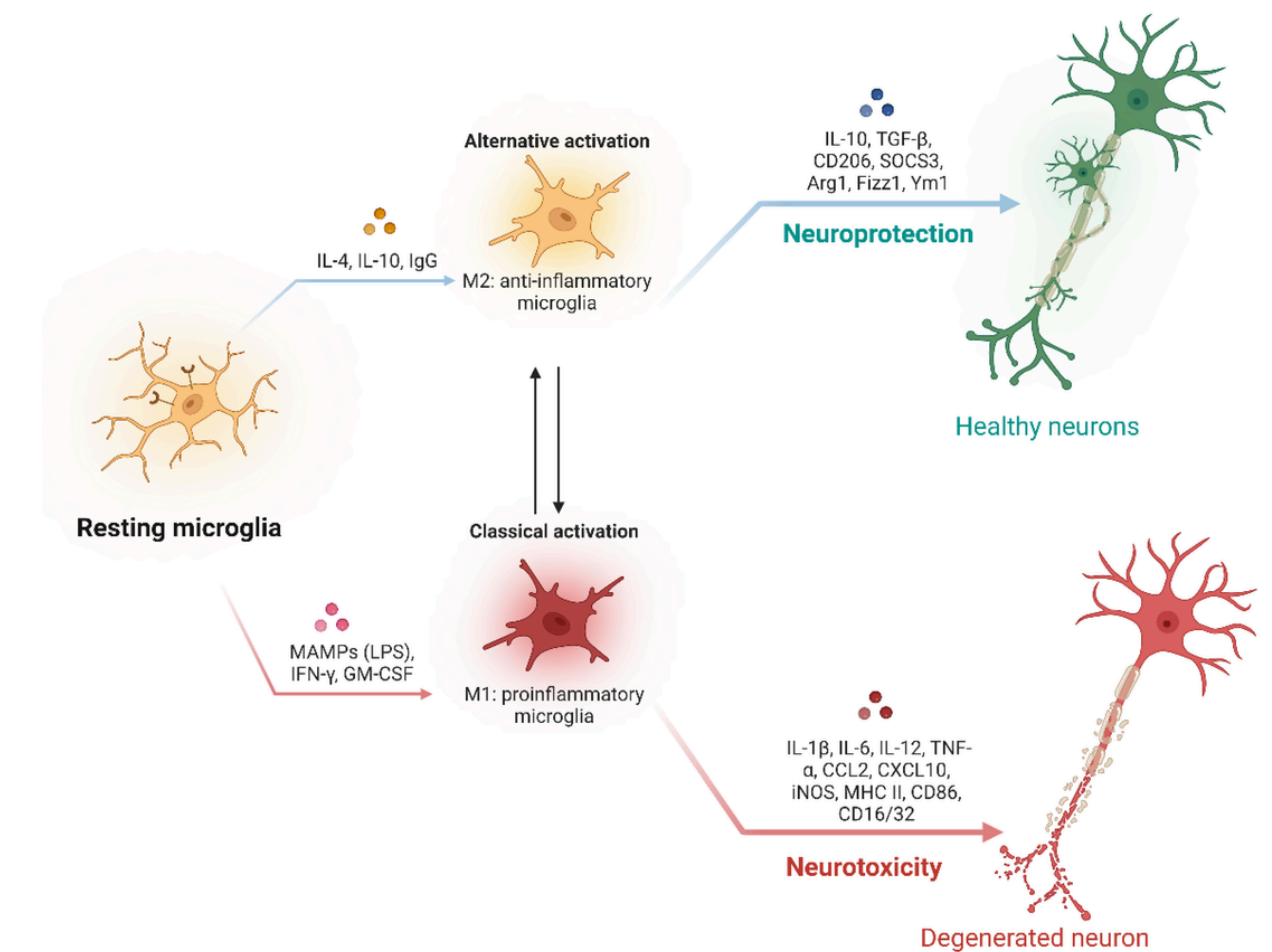
- An autoimmune disease that affects the brain and spinal cord.
- The neurons in the CNS are coated in myelin.
- This is supposed to insulate neurons for better signaling.
- For MS patients, the immune system attacks the myelin.
- This is called **demyelination**.



GLIAL ACTIVATION

MS - Multiple Sclerosis

- There are cells in the CNS called glial cells
- These are not neurons.
- Glial Activation = Resting state → Reactive state
- When this happens chemokines and cytokines are released
 - This recruits more immune cells
 - Inflammatory response is triggered.

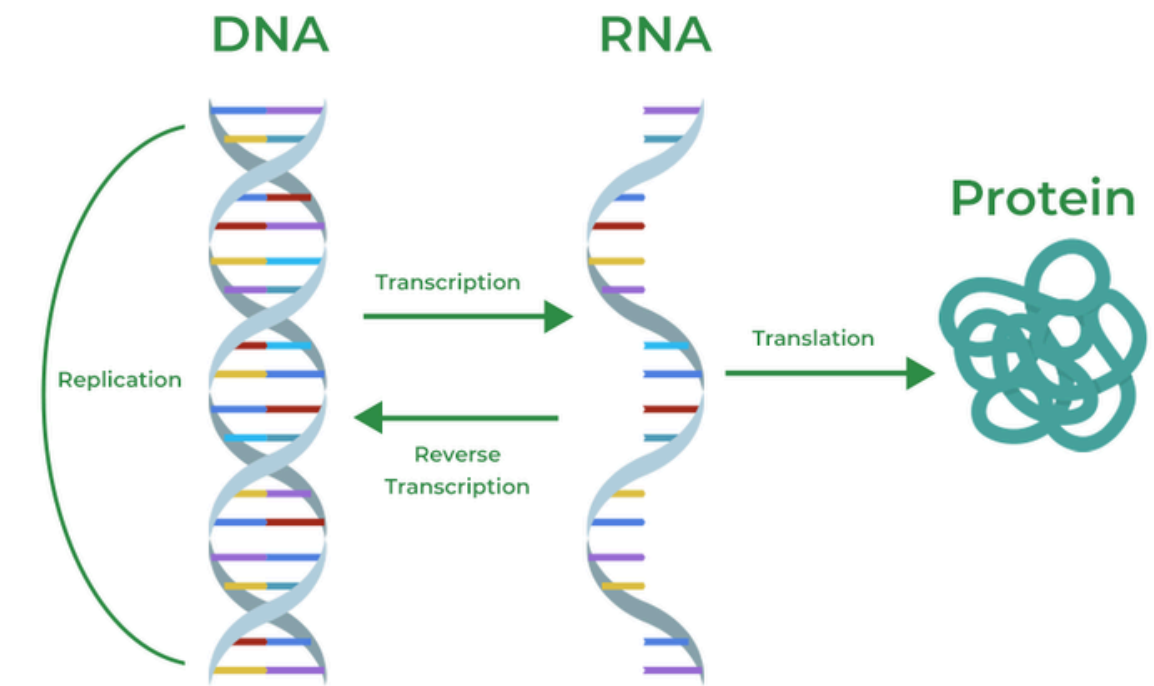


AIM

- Find out if upper respiratory infections can trigger autoimmune responses in the CNS.
- Find out the mechanism by which this happens.
- Find out the differentially expressed genes between pre- and post infection.

GENE → RNA → PROTEINS

- Specific **sections** of DNA are called **genes**, and these genes carry the code for specific traits.
- **Genes** switch **on** and **off**.
- How often they switch **on** and **off** is called **expression**
- More RNA = More gene expression



WHAT IS RNA SEQ?

RNAseq is a technique to measure expression for every single gene for a single cell or tissue.

gene	ctrl_1	ctrl_2	exp_1	exp_1
geneA	10	11	56	45
geneB	0	0	128	54
geneC	42	41	59	41

WHAT ARE WE TRYING TO DO?

- We have RNA seq data for infected mice and healthy mice. We are trying to find out **which genes are expressed differentially**.

gene	ctrl_1	ctrl_2	exp_1	exp_1
geneA	10	11	56	45
geneB	0	0	128	54
geneC	42	41	59	41

WHAT ARE WE TRYING TO DO?

Which gene is expressed differentially here?

gene	ctrl_1	ctrl_2	exp_1	exp_1
geneA	10	11	56	45
geneB	0	0	128	54
geneC	42	41	59	41

WHAT ARE WE TRYING TO DO?

geneB!

gene	ctrl_1	ctrl_2	exp_1	exp_1
geneA	10	11	56	45
geneB	0	0	128	54
geneC	42	41	59	41

WHICH GENE IS DIFFERENTIALLY EXPRESSED?

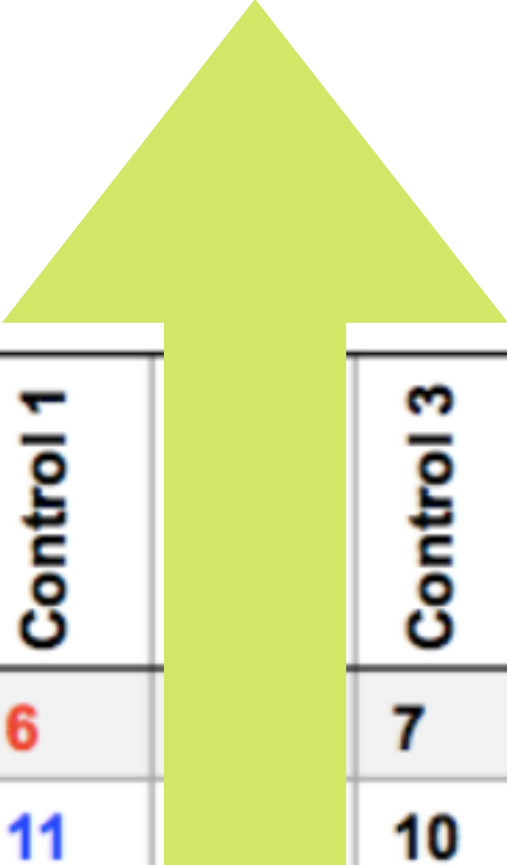
	Control 1	Control 2	Control 3	Sample 1	Sample 2	Sample 3
Gene A	6	5	7	17	10	11
Gene B	11	11	10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

WHICH GENE IS DIFFERENTIALLY EXPRESSED?

	Control 1	Control 2	Control 3	Sample 1	Sample 2	Sample 3
Gene A	6	5	7	17	10	11
Gene B	11	11	10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

WHY?

Very Big Numbers



	Control 1		Control 3	Sample 1	Sample 2	Sample 3
Gene A	6		7	17	10	11
Gene B	11		10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

WHY?

Very Big Numbers Very Small Numbers

	Control 1	Control 3	Sample 1	Sample 3
Gene A	6	7	17	11
Gene B	11	10	3	2
Gene C	200	150	50	1
Gene D	0	1	2	0

WHICH GENE IS DIFFERENTIALLY EXPRESSED?

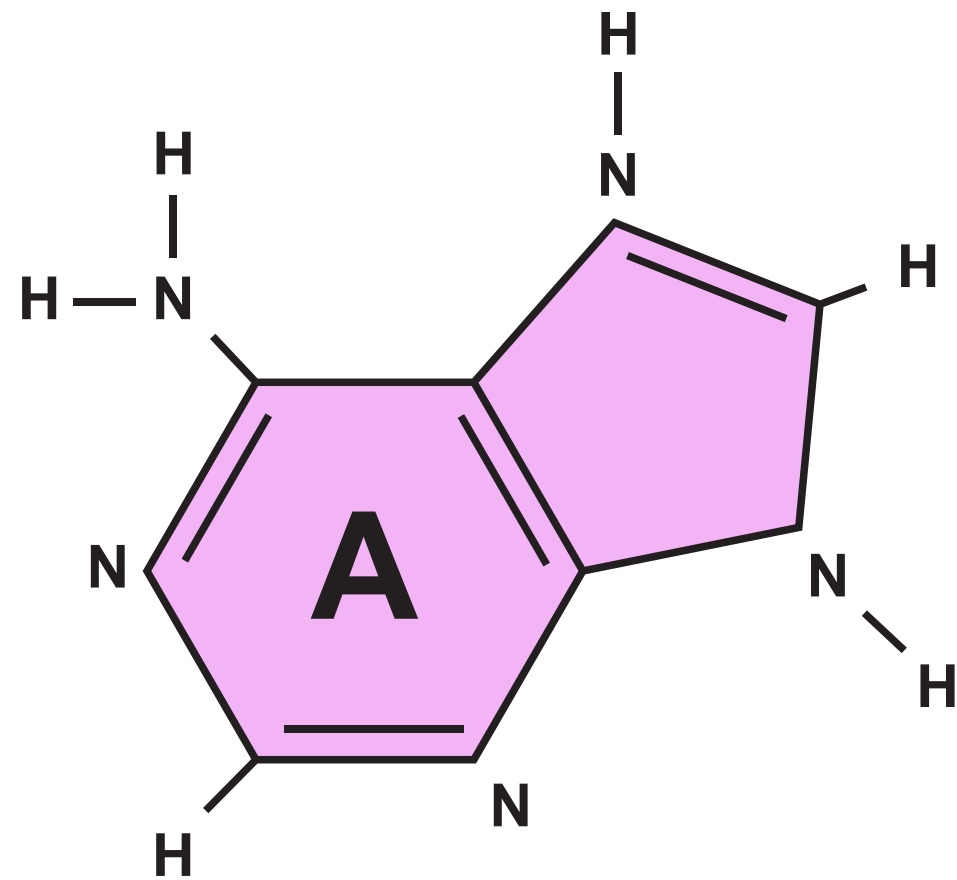
	Healthy 1	Healthy 2	Infected 1	Infected 2
AT1G01010	5	13	15	15
AT1G01020	182	177	121	144
AT1G01030	16	11	9	10
AT1G01040	30	32	113	132
AT1G01046	0	1	0	2
AT1G01050	87	82	114	107

WHICH GENE IS DIFFERENTIALLY EXPRESSED?

	Healthy 1	Healthy 2	Infected 1	Infected 2
AT1G01010	5	13	15	15
AT1G01020	182	177	121	144
AT1G01030	16	11	9	10
AT1G01040	30	32	113	132
AT1G01046	0	1	0	2
AT1G01050	87	82	114	107

WE HAVE ~20,000 GENES

So identifying the differentially expressed genes becomes annoyingly painful.



WHAT CAN WE DO?

Statistics & R!

RESULTS

Influenza A Infection Causes EAE Onset in 2D2 Mice

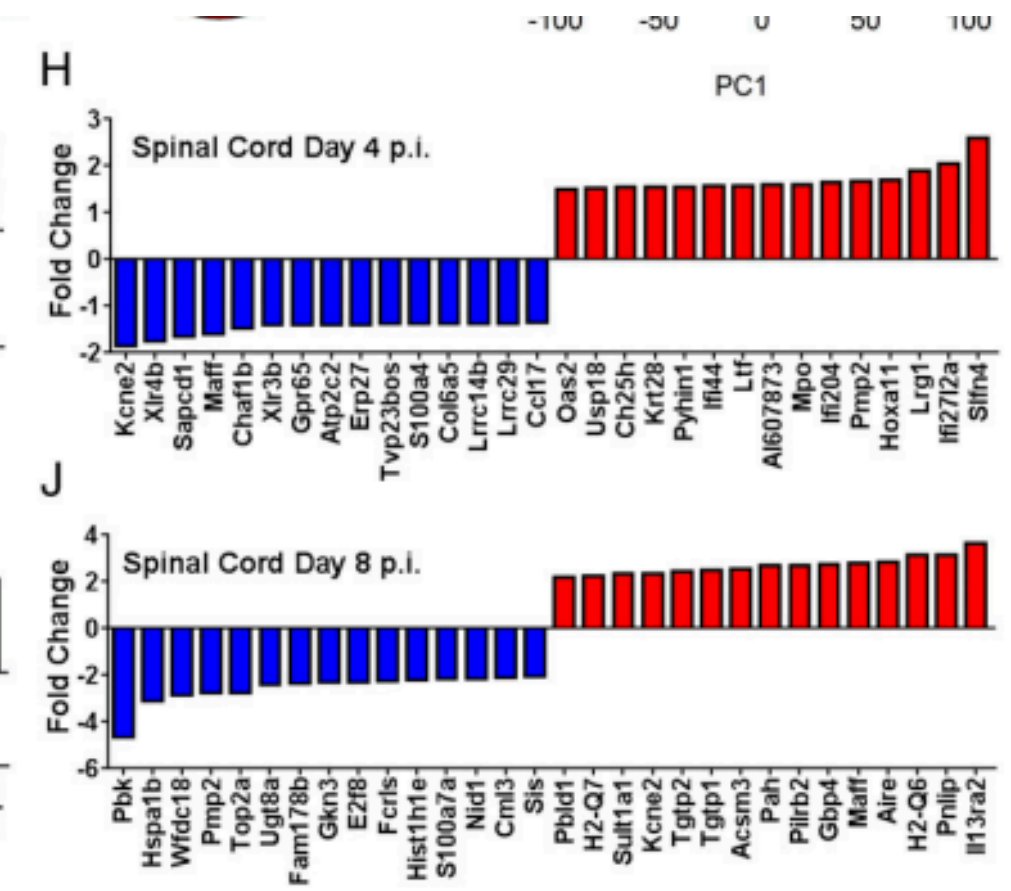
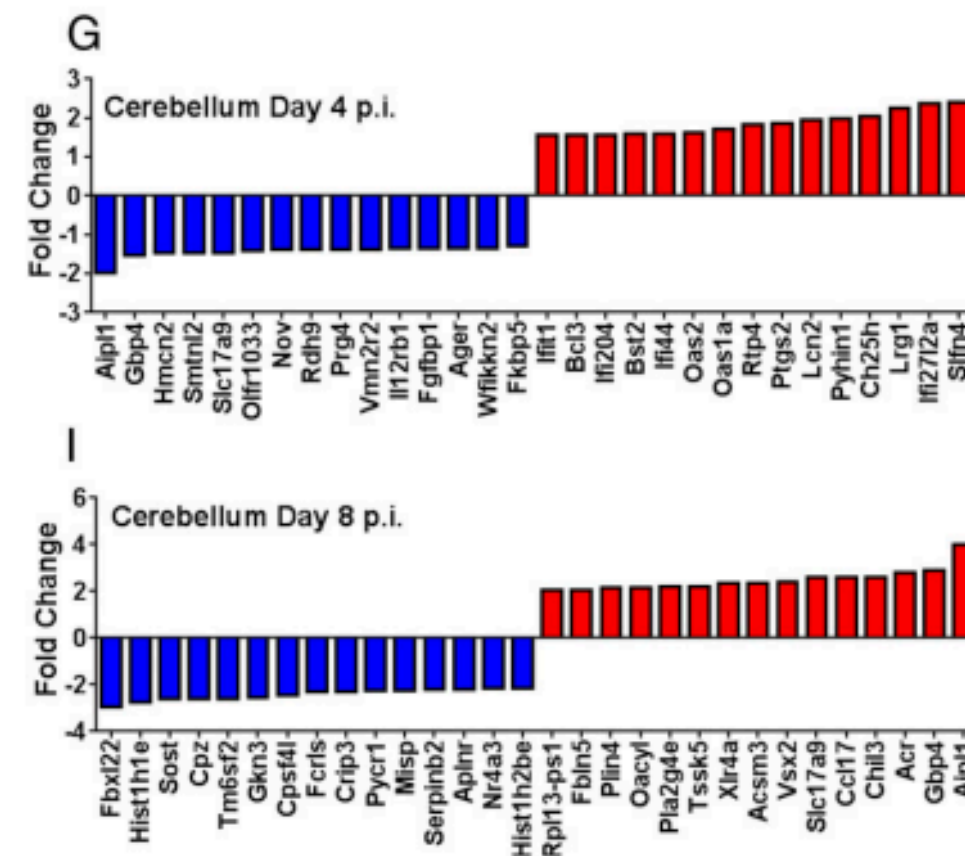
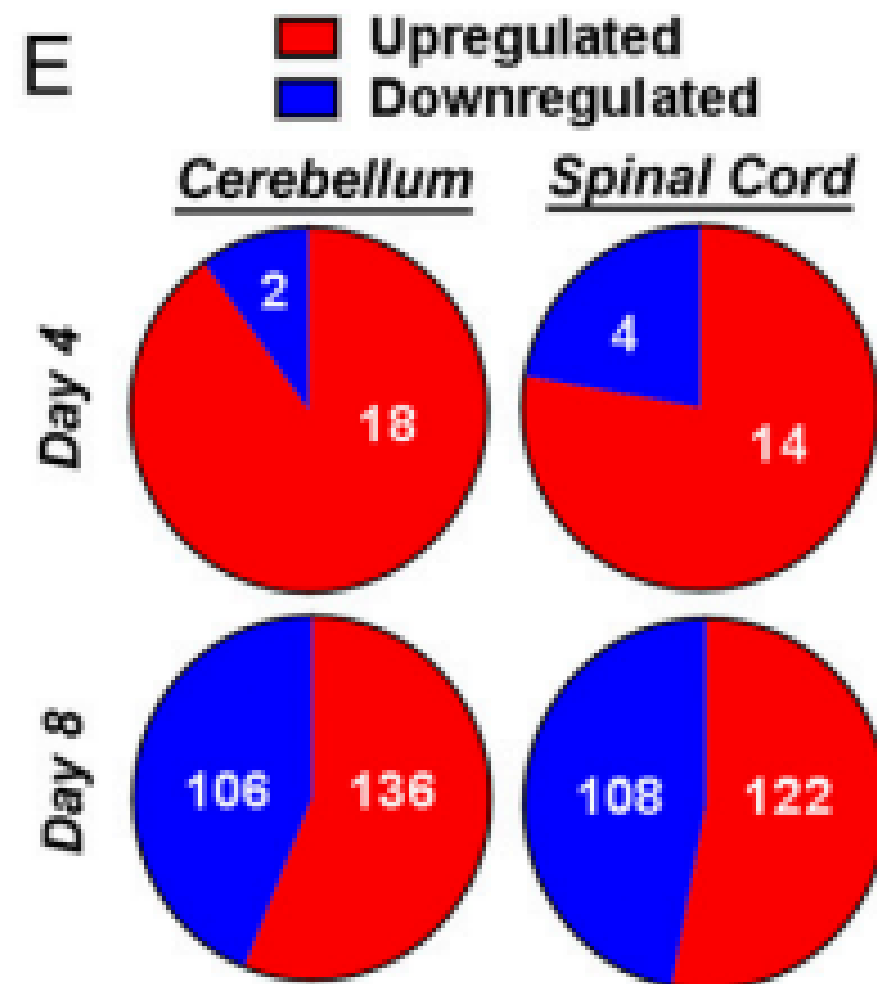
- 29% of 2D2 mice developed signs of EAE within 2 weeks of infection.
- However, the control mice did not develop signs of disease
- There is confirmed demyelination in the spinal cord and the cerebellum of influenza infected mice.

Table 1. Effect of influenza infection on neurological signs of EAE

Genotype	Inoculum	Incidence	Percent	Maximum score*	Day of onset*
C57BL/6	FLU	0/5	0	NA	NA
2D2	Saline	0/13	0	NA	NA
2D2	FLU	7/24	29	2	10

RESULTS

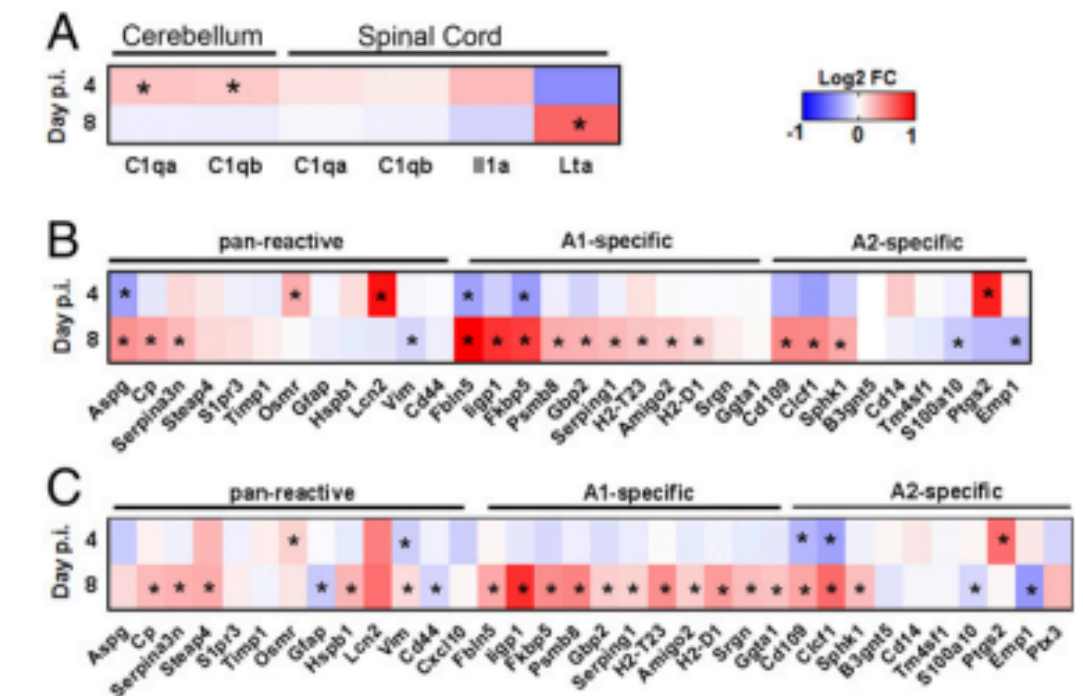
Infection causes alternations to cerebellar and spinal cord transcriptome.



RESULTS

Infection Up-regulates genes associated with glial activation

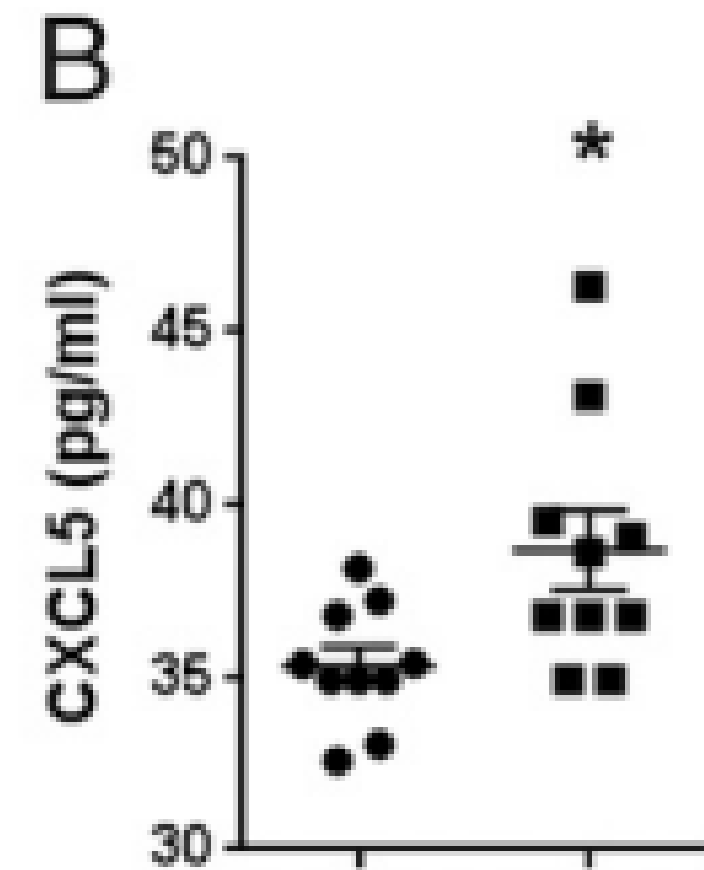
- Infection induces an “A1” reactive astrocyte phenotype.
- This is triggered by microglial activation and production of certain chemokines.
- Predictors of A1-reactive astrocyte phenotype were elevated in both cerebellar and spinal cord tissues.
- Hence, upper respiratory infections trigger reactive gliosis.



RESULTS

Infection causes immune cell surveillance of the CNS

- Influenza Infection causes certain genes to be upregulated.
- This leads to increased levels of cytokines.
- This recruits immune cells such as monocytes and neutrophils.
- So elevated levels CXCL5, attracted monocytes and neutrophils.
 - This is beleived to trigger an MS relapse



METHODS

RNA-seq

- Used to get the expression of each gene.
- 2 separate experiments were carried out.

Flow Cytometry

- This is a laser based technique to identify chemical properties of cells.

CONCLUSION

- Influenza Infections are capable of causing relapses in patients affected by autoimmune diseases affecting the CNS.
- This is evident by the glial activation that causes immune surveillance of the CNS resulting in neuroinflammation.
- Moreover, up-regulation of genes related to IFN- γ signalling contributed to relapse.

LETS DISCUSS FUTURE APPLICATION OF THIS STUDY

WHAT ARE THE LIMITATIONS OF THE STUDY?